PROGRAM OVERVIEW

Cap Collective

Turning Bottle Caps into Community Impact

"Join the Bridge. Create the Impact."

Overview · Generated 2026-10-10

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Cap Collective — Program Overview

> *"The distance between an idea and its impact is the bridge we build today."*

What Is Cap Collective?

Cap Collective is a **student-led circular recycling program** that captures discarded plastic bottle caps and transforms them into useful products — while funding the school garden. Students run every part of it: collection, cleaning, shredding, melting, product design, sales, and impact reporting.

Bottle caps are a **recycling blind spot**. Mechanical recycling centers process large items like bottles, but caps are too small — they fall through sorting screens or jam high-speed equipment. The result: a perfectly good, high-grade plastic is thrown away, every day, in every school.

Cap Collective captures that resource **at the source** and turns it into value.

The Problem in One Line

Only about **3% of bottle caps** are recycled globally. The rest end up in landfill or the ocean — where they look like food to sea turtles and fish.

The Solution: A 5-Step Micro-Factory in Your School

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1. COLLECT   →  HDPE (#2) & PP (#5) caps dropped in campus boxes
2. CLEAN     →  Remove liners · wash · air-dry 24 hours
3. SHRED     →  Uniform 5–8 mm flakes
4. MELT      →  Heat press: HDPE 140°C · PP 170°C  (never above 200°C)
5. REFINE    →  Laser-cut · injection-mold · press into finished goods
```

Closed loop: When a product breaks, it goes right back to Step 3. 100% recyclable, forever.

Why Caps?

Property	Why It Matters
HDPE (#2) & PP (#5)	Melt predictably; safe when heated carefully
High density & color	Strong products with beautiful marbled finishes
Abundant	Every school produces thousands per term
Food-safe grade	Suitable for useful everyday items
Ignored by industry	An untapped resource, not a nuisance

No PVC (#3) is ever used — it releases toxic gas when heated.

What Students Make

Coasters · keychains · phone stands · desk organizers · plant pots · art tiles · and more.

Each product uses a known number of caps (e.g., a keychain ≈ 4 caps, a coaster ≈ 15 caps), so students can see — and calculate — exactly how waste becomes value.

Impact You Can Measure

The program is built around real, trackable numbers. Targets **per semester**:

Metric	Target
Caps collected	10,000+
Plastic diverted	35+ kg
CO_2e avoided	65+ kg
Product revenue	$500+
Student participation	80%

A live dashboard counts caps automatically (via an optional sensor) and shows the impact in real time.

Three Ways to Start (Any Budget)

Tier	Cost	Best For
Core	$500 – $1,000	Primary schools, clubs, pilots
FabLab	$2,500 – $5,000	Middle/high school makerspaces
Center	$10,000+	Technical schools, district hubs

Start small. A pilot with boxes, a wash station, and a heat press proves the concept. Expand from there.

Learn While You Build

Cap Collective is a **full curriculum platform**, not just a recycling bin. It spans four grade bands:

- **K-2 — Cap Detectives:** observation, sorting, storytelling
- **3-5 — Cap Engineers:** measurement, building, basic design
- **6-8 — Cap Innovators:** data, sensors, entrepreneurship
- **9-12 — Cap Systems Designers:** engineering, life-cycle analysis, policy, scale

Every lesson links to standards (NGSS, Common Core, IB) and comes with printable worksheets.

Safety Is Non-Negotiable

- **No certification = no machine access.**
- Maximum temperature: **200°C** — never exceeded.
- Ventilation mandatory for all heating.
- PPE required per station.
- Every machine has a Standard Operating Procedure and a student certification card.

The Complete Implementation Kit

This program ships with **29 documents** covering the entire journey — from convincing your principal to handing the program off to the next year's students:

Plan & Pitch · Readiness audit · Launch deck · Speaker notes · School buy-in & approvals **Fund & Buy** · Procurement, payment & finance · Grant & fundraising templates **Run It** · Teacher guide · Command Center sheets · Safety SOPs & risk assessment **Teach It** · Lesson plans · Student worksheets · Curriculum mapping · Certification & training **Launch It** · Launch-week run-of-show · Communications kit · Printable resources **Sell It** · Eco-Store operations · Product development & QC **Grow It** · Engagement & gamification · Accessibility & inclusion · Impact reporting **Scale It** · District replication · Continuity & handover · Data science & AI extensions

Plus: importable spreadsheets, parametric CAD starters, and sensor code.

How to Get Started

1. **Read** the Implementation Readiness Audit — an honest picture of what it takes.

2. **Pitch** using the Launch Deck to your school leaders.
3. **Approve & fund** with the buy-in and finance playbooks.
4. **Train** staff and students with the certification pack.
5. **Launch** with the run-of-show and communications kit.
6. **Measure & report** with the dashboard and reporting templates.
7. **Scale & sustain** using the district and continuity guides.

Our Guiding Idea

> *Join the Bridge. Create the Impact.*

Cap Collective proves that sustainability isn't something students learn about — it's something students **build**.

Cap Collective — Empowering Sustainability through Environmental Stewardship and Economic Innovation.